# Cryptography & Network Security

Midterm Preparation Guide | By: Professor

***Professor's Advice:*** *Assalam o Alaikum beta! Dekho, tension bilkul nahi leni. 40 saal se bacho ko yehi subjects parha raha hoon. Cryptography math aur logic ka khel hai. In notes ko carefully parho, formula samjho, aur slides ko ratta lagane ki bajaye step-by-step logic apply karo. Mid asani se pass ho jayega!*

## Week 1: Classical Ciphers (Caesar & Playfair)

**Concept:** Shuruati weeks mein hum basics parhte hain. Encryption ka matlab hai simple text (Plaintext) ko aisi form (Ciphertext) mein badalna jo kisi ko samajh na aaye. Decryption iska ulta process hai.

### 1. Caesar Cipher (Shift Cipher)

Yeh sab se basic cipher hai. Is mein hum alphabet ke letters ko kisi fix number (Key) se aage shift kar dete hain.

- **Encryption Formula:** `C = (P + K) mod 26`
- **Decryption Formula:** `P = (C - K) mod 26`

*Example:* Agar Plaintext (P) = 'HELLO' aur Key (K) = 3 hai.
H (7) + 3 = 10 (K)
E (4) + 3 = 7 (H)
Result: Ciphertext = 'KHOOR'

### 2. Playfair Cipher

Is cipher mein 5x5 ka matrix banta hai. Hum letters ko pairs (do do ke joray) mein divide karte hain. Rules yaad rakhne wale hain:

1. Agar dono letters same row mein hain: right side wale letters le lo.
2. Agar dono letters same column mein hain: neechay (bottom) wale letters le lo.
3. Agar rectangle ban raha hai: unke opposite corners wale letters le lo.

📺 **YouTube Recommend:** Hindi/Urdu mein asaan explanation ke liye YouTube par search karein: "Playfair cipher Gate Smashers Hindi" (Bohat achi examples hain is channel par).

# Missed Slide Topics: Fundamentals & Math

**Security Architecture:** Slides ke shuru mein hamesha security concepts hote hain jinhein yaad rakhna lazmi hai, kiyunke mcqs inhi mein se bante hain.

## 1. CIA Triad & Attacks

- **Confidentiality:** Data secret rahe aur unauthorized log na parh sakein (Encryption se achieve hoti hai).
- **Integrity:** Data raste mein change (modify) na ho (Hashing / MAC se achieve hoti hai).
- **Availability:** System hamesha authorized users ke liye available rahe.
- **Passive Attack:** Attacker sirf data ko read karta hai (e.g., Eavesdropping, Traffic Analysis). Isay pakarna mushkil hai.
- **Active Attack:** Attacker data ko change karta hai ya system ko rokta hai (e.g., DoS, Modification, Masquerade). Isko detect karna asaan hai.

## 2. Mathematics for Crypto (Euclidean Algorithm)

Slides mein Modulo arithmetic aur GCD (Greatest Common Divisor) nikalne ke numericals lazmi hote hain. Euclidean algorithm ka basic rule yeh hai:

$$GCD(A, B) = GCD(B, A \bmod B)$$

Jab tak remainder (A mod B) zero na aa jaye, yeh step bar bar repeat karte hain. Yeh agay chal kar Asymmetric crypto ki foundation banti hai.

## 3. Vigenère Cipher (Polyalphabetic)

Caesar cipher weak tha kiyunke usme ek hi key (shift) thi. Vigenère mein ek keyword (key) use hota hai, aur har letter ke liye shift alag alag change hota hai. **Formula:**

$$C_i = (P_i + K_i) \bmod 26$$

## Week 2: Symmetric Encryption & DES

**Concept:** Symmetric key ka matlab hai Encryption aur Decryption dono ke liye ek hi key use hoti hai. DES (Data Encryption Standard) ek Block Cipher hai, yaani yeh data ko blocks (64-bits) mein divide kar ke encrypt karta hai.

### DES (Data Encryption Standard) Working Logic

Beta yaad rakho, DES Feistel structure par based hai. Iski total **16 rounds** hoti hain. Key ka size 64-bit hota hai lekin har round mein sirf 48-bit key use hoti hai (16 bits parity check ke liye drop ho jate hain).

| Step Name | Kiya hota hai? (Roman Urdu) |
|---|---|
| **Initial Permutation (IP)** | 64-bit block aata hai aur uske bits ki jagah (position) shuru mein sharam-o-haya ke baghair aage peeche (shuffle) kar di jati hai. |
| **Splitting** | 64-bit data ko 2 hisso mein divide karte hain: Left (32-bit) L0 aur Right (32-bit) R0. |
| **Feistel Round (1-16)** | R(i-1) ko Expansion permutation se 48-bit banaya jata hai. Phir Round Key ke sath XOR hota hai. |
| **S-Boxes (Substitution)** | Yeh sab se important step hai. 48-bit data S-Boxes mein jata hai aur wapas shrink ho kar 32-bit ban jata hai. (Yeh confusion create karta hai). |
| **Final Permutation** | 16 rounds ke baad Left aur Right ko jor kar dobara swap karte hain aur Final Permutation se guzarte hain (Ciphertext ready!). |

**Exam Tip:** Paper mein poora 16 round ka math solve karne ko nahi aata. Aap se sirf S-Box ka size, Initial Permutation ki logic, aur Feistel Network ka diagram banwaya jayega.

📺 **YouTube Recommend:** DES ka Feistel structure samajhne ke liye Neso Academy best hai:
"DES encryption Neso Academy Hindi"

# Week 3: AES (Advanced Encryption Standard)

**Concept:** DES purana aur weak ho gaya tha, is liye AES laya gaya. Yeh Feistel structure nahi hai, balkeh Substitution-Permutation Network (SPN) par kaam karta hai. Is mein data ko bytes ki form mein 4x4 matrix (jise State kehte hain) mein process kiya jata hai.

## AES ki Basics aur Key Sizes

AES mein block size hamesha **128-bits** (16 bytes) hota hai. Lekin Key ke sizes mukhtalif ho sakte hain:

- 128-bit Key = 10 Rounds
- 192-bit Key = 12 Rounds
- 256-bit Key = 14 Rounds

### AES Round Transformations (Steps)

Har AES round (akhri round ke ilawa) mein yeh 4 steps hotay hain. Inko sequence mein yaad rakhna hai:

1. **SubBytes:** Har byte ko ek table (S-Box) se replace karte hain. (Non-linearity ke liye)
2. **ShiftRows:** Matrix ki rows ko circularly shift karte hain. (1st row shift nahi hoti, 2nd ek dafa, 3rd do dafa...)
3. **MixColumns:** Har column ke data ko math ke matrix multiplication se mix kiya jata hai. (Yeh diffusion ke liye hota hai)
4. **AddRoundKey:** Jo block ab tak bana hai, usko is round ki key (Sub-key) ke sath XOR kar diya jata hai.

*Note:* Akhri (Last) round mein MixColumns ka step nahi hota!

📺 **YouTube Recommend:** AES ke steps easily samajhne ke liye search:
"AES encryption Gate Smashers" (Isme SubBytes aur ShiftRows ki example board par bohat achi di hai).

# Missed Slide Topics: Block Cipher Modes of Operation

**Concept:** Jab plain data ek block (jaise 64-bit ya 128-bit) se bara ho, to usay multiple blocks mein divide kiya jata hai. In blocks ko aik sath kaisay encrypt karna hai, isko **Modes of Operation** kehte hain. Slides mein yeh topic DES/AES ke as pas lazmi aata hai.

## 1. ECB (Electronic Codebook) Mode

Yeh sab se basic aur weak mode hai. Is mein har block alag alag (independently) encrypt hota hai.

- **Faida:** Fast hai, parallel processing easily ho sakti hai.
- **Nuqsan:** Agar 2 plaintext blocks same hain, to unka ciphertext block bhi hamesha same banega. Attacker pattern pehchan sakta hai.

## 2. CBC (Cipher Block Chaining) Mode

Slides mein is mode par numericals aur diagram ka zikr hota hai. Is mode mein har block ko encrypt karne se pehle pichle ciphertext block ke sath **XOR** kiya jata hai.

| Step | CBC Mode Working Logic (Roman Urdu) |
|---|---|
| **First Block (IV)** | Pehle block ke liye pichla ciphertext nahi hota, is liye ek random temporary block jise **IV (Initialization Vector)** kehte hain, XOR ke liye use hota hai. |
| **Encryption Rule** | $C_i = E(K, P_i \oplus C_{i-1})$<br>*(Plaintext ko pichle Ciphertext se XOR kar ke Encrypt karo)* |
| **Decryption Rule** | $P_i = D(K, C_i) \oplus C_{i-1}$ |
| **Nuqsan** | Parallel encryption nahi ho sakti kiyunke agla block pichle block ka wait karta hai. |

📺 **YouTube Recommend:** Modes of operation ki diagrams samajhne ke liye search karein:
"Block Cipher Modes of Operation Gate Smashers Hindi"

## Week 4: RC4 Stream Cipher (From Books)

**Concept:** Beta, jahan DES aur AES "Block Ciphers" thay, **RC4 ek Stream Cipher hai**. Iska matlab yeh data ko block by block nahi, balkeh bit by bit ya byte by byte encrypt karta hai. Slides mein aksar yeh miss ho jata hai is liye book ki proper theory se explanation de raha hoon.

RC4 ka basic kaam ek "Keystream" generate karna hai (jo ke pseudo-random bytes hote hain). Phir hum Plaintext ko is Keystream ke sath sirf XOR (⊕) karte hain. Keystream generate karne ke 2 main steps hain:

### 1. KSA (Key-Scheduling Algorithm)

Is step mein hum ek Array 'S' banate hain jiski length 256 hoti hai (0 se 255 tak). Is array ko hum initial Key ki madad se completely shuffle (mix) kar dete hain.

```
      Initialization: For i = 0 to 255: S[i] = i T[i] = Key[i mod KeyLength]
Scrambling (Shuffle): j = 0 For i = 0 to 255: j = (j + S[i] + T[i]) mod 256 Swap(
S[i], S[j] )
```

### 2. PRGA (Pseudo-Random Generation Algorithm)

Ek dafa S-array mix ho gaya, ab hum plaintext ki length jitni lambi random stream generate karte hain.

```
      Generation loop: i = 0, j = 0 While(more bytes to encrypt): i = (i + 1) mod
256 j = (j + S[i]) mod 256 Swap( S[i], S[j] ) t = (S[i] + S[j]) mod 256
Keystream_Byte (K) = S[t] Ciphertext = Plaintext_Byte ⊕ K
```

**Professor's Tip:** Book mein yeh algorithmic form mein aata hai. RC4 ki khasiyat yeh hai ke yeh software mein implement karna intehai asaan aur fast hai, is liye yeh pehle WEP (Wi-Fi) waghera mein use hota tha, magar ab is mein security vulnerabilities (weaknesses) daryaft ho chuki hain, is liye modern systems isay avoid karte hain.

📺 **YouTube Recommend:** RC4 algorithm dry run search:
"RC4 encryption algorithm step by step in Hindi Gate Smashers"

## Midterm Practice Questions & Final Advice

Beta, exam se pehle in sawalat ko pen aur paper le kar khud solve lazmi karna. Sirf parhne se lagta hai ke aata hai, magar exam mein calculations galat ho jati hain.

### Expected Numerical & Theoretical Questions:

1. **Basic Math (GCD):** Euclidean Algorithm use kar ke GCD(105, 30) find karein. *(Answer: 15)*
2. **Caesar Cipher Practice:** Plaintext = "UNIVERSITY", Key = 5. Ise encrypt karein. Phir jo Ciphertext aaye, usay wapas Decrypt kar ke verify karein.
3. **Vigenère Cipher:** Plaintext = "HELLO", Keyword = "KEY". Ciphertext banayein (Modulo 26 mathematics step-by-step likhein).
4. **Playfair Matrix Generation:** Keyword = "SECURITY" (Yaad rakho I aur J aik hi box mein aate hain). Is matrix ko bana kar "HELLO" ko encrypt karein.
5. **Block Cipher Modes:** ECB aur CBC mein main farq kya hai? IV (Initialization Vector) ka kaam sirf CBC mein kyu zaroori hota hai?
6. **DES Architecture:** Feistel network ka diagram draw karein. Agar ek block 64-bit ka hai toh Initial Permutation ke baad Left(L0) aur Right(R0) ka size kya hoga? *(Answer: 32-bits each)*
7. **AES Transforms:** AES mein SubBytes aur ShiftRows step ka main maqsad (purpose) kya hai? *(Answer: SubBytes confusion create karta hai, ShiftRows diffusion laata hai)*
8. **RC4 Stream Cipher:** RC4 mein KSA aur PRGA ka farq batayein? Book ki context mein Array S ki initial range kitni hoti hai? *(Answer: S array 0 se 255 tak hoti hai, total 256 bytes)*.

> ***Last Word from Professor:*** *Mujhe tum par poora yakeen hai. Raat ko time par sona, exam mein calculations dhyan se karna (khusoosan modulo arithmetic). RC4 wala part jo slides mein nahi tha, wo upar step-by-step likh diya hai usko 2 dafa parh lena. Good luck bacho! Allah kamyab kare.*